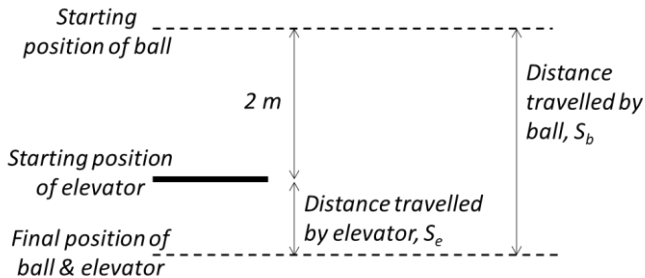


back at the starting point.

3

D



$$S_b - S_e = 2$$

$$2.0 = [ut + (0.5)(9.81)t^2] - [ut + (0.5)(5.8)t^2]$$

$$t = 1.00 \text{ s}$$

Or recognise that the effective acceleration = $9.81 - 5.8 = 4.01 \text{ m s}^{-2}$.

Using $s = ut + \frac{1}{2}at^2$, $2.0 = \frac{1}{2}(4.01)t^2$, giving $t = 1.00 \text{ s}$.

4

A

The question asks for the HORIZONTAL forces only.